

Paper 2 theory audit: three questions, answered from the papers in hand

Cowork note for Chris — HardDisks / hspist3

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How to read this

Every statement carries one tag. **[SOURCE X p. n]** means the paper says it, on that page or equation. **[DERIVATION]** means it follows by algebra from a sourced statement under assumptions I name. **[INFERENCE]** is our interpretation of the simulation. **[DATA]** is a number CC measured. **[OPEN]** is not established and says what would establish it. Nothing below is tuned to the data; where the literature and the box disagree, the disagreement is kept and the violated assumption is named.

Units as always: $k_B T = m = \sigma = 1$, $\sigma = 2r$, compartment $L \times H = 39.25 \times 10$, $N_s = 50$ disks, $\eta = 0.100$, number density $n = N_s/(LH) = 0.1274$, travel $\Delta x = 3.93 \sigma$, so $\eta \rightarrow 0.1112$ and $L \rightarrow 35.32$.

1 What is in hand, and what each paper carries

Paper	System and assumptions	Carries, for us
Jarzynski 1997, PRL 78, 2690 (arXiv version attached)	finite classical system, <i>canonical</i> initial ensemble at T ; isolated Hamiltonian switching suffices (pp. 4–5, Eqs. 6–8)	the equality $\overline{e^{-\beta W}} = e^{-\beta \Delta F}$ and its exact initial-ensemble requirement; the word “dissipated work” for $\overline{W} - \Delta F$ (p. 2)
Jarzynski 1997, PRE 56, 5018 (master-equation version, attached)	same result for stochastic (bath-coupled) dynamics	not needed for Levels 0–5; keep for the Jarzynski experiment
Sun 2003, JCP (path sampling)	how to converge the exponential average	Jarzynski experiment only
Lua & Grosberg 2005, JPCB 109, 6805	<i>one</i> ideal-gas molecule, canonical velocity, piston <i>receding</i> at fixed v_p for fixed time $\tau = 1$ (p. 6806, Fig. 1)	exact $P(W)$ and the fast-piston limit for <i>expansion</i> ; the rare-event cost of Jarzynski (p. 6809)
Sivak & Crooks 2012, PRL 108, 150601 (attached)	system in contact with a bath, canonical; piston position named as a control parameter (p. 1)	definition of the <i>nonequilibrium free energy</i> $F = \langle E \rangle - S/\beta$ and its meaning as maximum extractable work (p. 1); linear response as an integrated force covariance (p. 3, Eqs. 10–11)
Sivak & Crooks 2012, PRL 108, 190602 (“Thermodynamic metrics”, <i>not</i> attached)	bath, canonical, slow protocol	the friction $\zeta(\lambda) = \beta \int_0^\infty \langle \delta X(0) \delta X(t) \rangle dt$ and $W_{\text{ex}} = \int \zeta \dot{\lambda}^2 dt$; quoted from memory, page numbers to be checked
Malek Mansour, Garcia & Baras 2006, PRE 73, 016121	2D hard disks, adiabatic piston, dilute ($n = 0.0028$), hydrodynamics vs MD	linear flow profile (Eq. 15), flow kinetic energy as renormalised mass (Eq. 18), <i>viscous</i> piston friction (Eq. 17), Enskog η, ζ, g_2 (Eqs. 8–11), damped-oscillator period and Q (Eqs. 33–39); fast mechanical vs slow thermal stage (p. 1)
Croignani, Di Porto & Conti, adiabatic piston and the second law	ideal gas, Langevin piston with <i>collisional</i> friction (Eq. 1)	the Brownian-piston wander around the maximum-entropy position (p. 2): this is Paper 1’s slow mode
Wu, Liu, Reese & Zhang 2016, JFM 794, 252	hard disks in a channel of width L/σ , generalised Enskog	wall layering in narrow channels; background for why $Z_{\text{wall}} > Z_{\text{bulk}}$ at $H = 10$; not used below
Kirkwood 1946, JCP 14, 180; Green 1954, JCP 22, 398; Kubo 1957, JPSJ 12, 570 (<i>not</i> attached)		the origin of the force-autocorrelation friction and its plateau problem; cited, not quoted

2 Question 1 — what is ζ in a closed box?

2.1 The formula and what it assumes

Sivak–Crooks (190602) define, for a control parameter λ with conjugate force $X = -\partial E/\partial \lambda$,

$$\zeta(\lambda) = \beta \int_0^\infty \langle \delta X(0) \delta X(t) \rangle_\lambda dt, \quad W_{\text{ex}} \simeq \int \zeta(\lambda) \dot{\lambda}^2 dt, \quad (1)$$

for a system in contact with a bath, in canonical equilibrium at fixed λ , driven slowly compared with the decay of the force fluctuations. The attached Letter (150601, p. 3, Eqs. 10–11) shows the same structure one level down: the linear response of any observable to a protocol is $\beta \int \dot{\lambda} \langle \delta \mathcal{G} \delta B \rangle dt'$, and the boundary term vanishes because “for an ergodic system all measurements separated by infinite time are uncorrelated”. **[SOURCE S–C 150601 p. 3]**

Three assumptions are therefore built in: (a) canonical preparation with a bath; (b) the covariance decays, so the integral to ∞ exists and is reached on a time short compared with the protocol; (c) $\dot{\lambda}$ slow on that same time. The plateau problem is older than all of this: Kirkwood 1946 already noted

that for a Brownian particle the running integral must be read on an intermediate plateau, after the collisional decay and before the finite-system recurrences. [\[SOURCE Kirkwood 1946, cited\]](#)

2.2 What the box does

[\[DATA\]](#) The running integral on the held wall at $\eta = 0.10$: $2.55 \rightarrow 1.22$ (10σ) $\rightarrow 0.15$ (20σ) $\rightarrow -0.05$ (40σ), zero crossing at 19.8 – 21.5 σ -time, recurrence ≈ 39.5 , and at the compressed end the crossing moves to 18.0 (predicted 15.9 from L/c_s). Averaged past the first crossing: -0.07 . Measured slope $d\langle W \rangle/du$ for $u \leq 0.05$: -0.50 ± 0.67 .

2.3 Why an undamped cavity mode contributes nothing

[\[DERIVATION\]](#) A closed compartment with rigid walls has standing acoustic modes with $f_k = kc_s/(2L)$. If the wall force were carried by the fundamental alone,

$$C_{FF}(t) \propto \cos\left(\frac{\pi c_s t}{L}\right) \implies \int_0^\tau C_{FF} dt \propto \sin\left(\frac{\pi c_s \tau}{L}\right), \quad (2)$$

which is maximal at $\tau = L/2c_s$, crosses zero at $\tau = L/c_s$ and recurs with $2L/c_s$: 19.8 and 39.5 σ -time here. That is CC's curve. An undamped mode therefore contributes *zero* to ζ ; only its damping survives the integral. This is a first-mode model, consistent with the timings, not an exact expression for C_{FF} (ChatGPT's caveat, which I accept). [\[INFERENCE\]](#) for the identification with the data.

2.4 Two frictions, both in the literature, thirty times apart

Collisional (short-time). For an infinite-mass wall moving at u into a Maxwellian gas the momentum flux gives, to first order in u , [\[DERIVATION\]](#)

$$\zeta_0 = h n \sqrt{\frac{8mk_B T}{\pi}} g_2(\eta), \quad g_2 = \frac{1 - 7\eta/16}{(1 - \eta)^2} \quad \text{[SOURCE Mansour Eq. 11]}, \quad (3)$$

so here $\zeta_0 = 10 \times 0.1274 \times 1.596 \times 1.181 = 2.40$. CC's zero-lag value 2.53 is this to 5%. The same coefficient is the friction term of the Langevin adiabatic piston of Crosignani et al., Eq. (1) [\[SOURCE Crosignani p. 2\]](#), where it is correct because the piston's Brownian jitter is fast compared with L/c_s : the gas cannot co-move, the wall sees molecules at rest.

Viscous (zero-frequency). Mansour et al. assume a linear fluid velocity profile $v(x) = v_p x/X_p$ (Eq. 15) and derive the piston equation

$$\hat{M} \ddot{X}_p = L_y(\bar{P}_L - \bar{P}_R) - L_y v_p \left(\frac{\bar{\Gamma}_L}{X_p} + \frac{\bar{\Gamma}_R}{L_x - X_p} \right), \quad \Gamma = \zeta_{\text{bulk}} + \eta_{\text{shear}}, \quad (4)$$

[\[SOURCE Mansour Eq. 17\]](#) so the friction on a wall that compresses one compartment of length X_p is $\zeta_{\text{hyd}} = L_y \Gamma / X_p$. With their Enskog expressions (Eqs. 8–9) at $\eta = 0.100$: $\eta_{\text{shear}} = 0.314$, $\zeta_{\text{bulk}} = 0.017$, $\Gamma = 0.331$, [\[DERIVATION\]](#)

$$\zeta_{\text{hyd}} = \frac{10 \times 0.331}{39.25} = 0.084 \text{ (start)}, \quad 0.096 \text{ (end, } \eta = 0.1112), \quad \frac{\zeta_{\text{hyd}}}{\zeta_0} \approx 0.035. \quad (5)$$

The prediction for the slow end of Level 2 is then $W_{\text{diss}} = \bar{\zeta}_{\text{hyd}} u \Delta x \approx 0.09 \times 3.93 u = 0.35 u$: a slope of $+0.35$ against the measured -0.50 ± 0.67 . Consistent, and unresolvable with the present seeds; at $u = 0.05$ it is $0.02 k_B T$ against a quadratic term of $0.06 k_B T$, at $u = 0.2$ it is 0.07 against 1.0 .

What to write

The force memory on a wall of a closed compartment is not Markovian: its running integral is the Enskog collisional friction $\zeta_0 \approx 2.4$ at zero lag, is cancelled by the returning sound wave at L/c_s , recurs at $2L/c_s$, and its zero-frequency limit is the hydrodynamic viscous friction $\zeta_{\text{hyd}} = L_y \Gamma / X_p \approx 0.09$ (Malek Mansour et al. 2006, Eq. 17), thirty times smaller. The Sivak–Crooks slow-end term $\zeta u \Delta x$ is therefore predicted at $0.35 u$ and is below the resolution of 760 trajectories; the data show no linear term at the 0.7 level. “Friction is zero” is not the statement; “the Markovian friction is the viscous one and it is small” is.

2.5 A free cross-check against Paper 1

[DERIVATION] Mansour et al. linearise the same equation for a heavy piston between two compartments (Eqs. 33–39): a damped oscillator with, in their scaled time, $\omega_0 = 4$ and damping $\beta = 2\mu$, $\mu = \Gamma L_y \sqrt{2/(\hat{M}N)}$, $\hat{M} = M + mN/3$. Hence $Q = \omega_0/2\beta = 1/\mu$. For Paper 1’s A1 geometry at $\eta = 0.10$ ($N = 100$, $H = 10$, $\Gamma = 0.331$): $M = 1000$ gives $\mu = 0.0146$, $Q \approx 69$, amplitude loss $1 - e^{-\pi/Q} \approx 4.5\%$ per period, linewidth $\Delta f/f = 1/Q \approx 1.5\%$; $M = 100$ gives $Q \approx 25$ and 4% linewidth, but there $K \approx 0.86$ and their homogeneous-density assumption (valid for $K \ll 1$, $\alpha \gg 1$) is marginal. **[OPEN]** Compare $1/Q$ with the measured width of the position-spectrum peak in the A1 data; at 200 periods the bin is 0.5%, so a 1.5% line spans three bins, which is what CC’s “median 3 distinct peak bins per 10 seeds” looks like. Their theory is dilute ($n = 0.0028$); using the Enskog Γ at $\eta = 0.10$ is an extrapolation they do not make. If it holds, Paper 1’s linewidths give Γ , and Γ gives Paper 2’s ζ_{hyd} with no new runs.

3 Question 2 — may $W - W_{\text{qs}}$ be called dissipation?

Jarzynski names $\bar{W} - \Delta F$ “the dissipated work, associated with the increase of entropy during an irreversible process” **[SOURCE Jarzynski PRL p. 2]**, for a system that has a reservoir and re-equilibrates after the switch. Sivak–Crooks define the nonequilibrium free energy $F = \langle E \rangle - S/\beta$ on any distribution and state that its excess over equilibrium “equals the maximum work that can be done on [a] mechanical system while the original system relaxes to equilibrium (also known as the exergy)” **[SOURCE S–C 150601 p. 1]**. Both statements are about the state *after* relaxation.

[INFERENCE] At piston stop the gas is not relaxed. Energy conservation (Level 0) says $W = \Delta E_{\text{gas}}$ exactly, and W_{qs} is the part that a reversible path would also have put in. The remainder is, at that instant, stored:

$$W - W_{\text{qs}} = E_{\text{flow}} + E_{\text{compress}} + E_{\text{transient}} + (\text{already thermalised}), \quad (6)$$

and only the last term is heat. The first term has a sourced size: the linear profile $v = ux/L$ **[SOURCE Mansour Eq. 15]** gives $E_{\text{flow}} = \frac{1}{2} \rho H \int_0^L v^2 dx = N_s m u^2 / 6$, which is exactly the $mN/3$ added to the piston mass in their Eq. 18 (with $N_s = N/2$). **[DERIVATION]** So a steady compression flow carries $8.3 u^2$, not the measured $25.2 u^2$; the other two thirds must be compressional energy and the two impulses of the step protocol. A single impulsive start radiates a pulse of pressure $\rho c_s u$ for a time L/c_s , i.e. energy $\rho c_s u \cdot H \cdot uL/c_s = N_s m u^2$ (equipartitioned between kinetic and compressional in a sound wave), and the stop does the same, so A can lie anywhere between $N_s m/6$ and about $2N_s m$ depending on how much of the launched energy is still in the gas at stop. **[INFERENCE]** The coincidence $A \approx N_s m/2$ is not an identification.

What to write

Level 2 measures the finite-rate *excess energy* of the compressed gas at piston stop, $E_{\text{ex}} = W - W_{\text{qs}} = Au^2$ with $A = 25.2$ and no resolvable linear term. It is not dissipation: at that instant it is flow and compressional energy of the closed cavity plus what the step protocol launched, and it becomes heat only as the cavity modes damp. The Sivak–Crooks exergy $F_{\text{neq}} - F_{\text{eq}}$ is the part that a mechanical degree of freedom could still capture — which is precisely what the spring of Level 3 measures.

The cheap decisive test: ramp against step

Same box, same Δx , $u = 0.05, 0.1, 0.2$, 60 seeds each, piston velocity ramped linearly over $2L/c_s \approx 40 \sigma$ -time at start and at stop instead of the step. Fit A_{ramp} . If $A_{\text{ramp}} \ll A_{\text{step}}$ the u^2 term is protocol-launched acoustic energy; if $A_{\text{ramp}} \approx A_{\text{step}}$, decompose at stop into $\frac{1}{2}m \sum_b N_b \bar{v}_{x,b}^2$ (flow), $\sum_b (\delta p_b)^2 A_b / (2\rho c_s^2)$ (compression) and the rest. The flow term has a prediction: $N_s m u^2 / 6$.

[OPEN] Whether “fully relaxed” exists for this box at finite N : Mansour et al. (p.1) note that the mechanical-equilibrium state is degenerate in the thermodynamic limit and that the final state depends on the order of $N \rightarrow \infty$ and $t \rightarrow \infty$; Crosignani et al. (p.2) find the piston never stops but wanders around the maximum-entropy position. Paper 1’s slow mode is that wander. For Level 2 this is harmless, since W is read at stop; for Level 4 (heat channel through the divider) it is the physics.

4 Question 3 — the fast limit at fixed Δx

4.1 What Lua–Grosberg actually solve

[SOURCE L–G p. 6806, Fig. 1] One molecule, $k_B T = m = 1$, canonical initial velocity, piston *receding* at v_p for a fixed time $\tau = 1$ (expansion, $L \rightarrow L + v_p$). Work per collision $-w_1 = -2(v - v_p)v_p$ (Eq. 8), after n collisions $-w_n = -2nvv_p + 2n^2v_p^2$ (Eq. 9); the identity $\langle e^W \rangle = (L + v_p)/L = Z_{\text{final}}/Z_{\text{initial}}$ is recovered exactly (Eq. 11). Fast limit $v_p \gg 1$: single bounce dominates, $P_{W>0} \simeq e^{-v_p^2/2}/(\sqrt{2\pi}Lv_p^2)$ and $\langle W \rangle \simeq 4P_{W>0}$ (Eqs. 21–22): the mean work is *exponentially small* because almost no molecule catches a receding piston, and the Jarzynski average is carried by the Maxwell tail; the number of runs needed grows like $\exp[mL^2/\tau^2 k_B T]$ (p. 6809).

4.2 Why none of that transfers to compression, and what does

[DERIVATION] For a piston *advancing* at $u \gg v_{\text{th}}$ over Δx , every particle in the swept strip is hit once (after the hit it moves at $2u - v > u$ and outruns the piston), so

$$N_{\text{hit}} \simeq N_s \frac{\Delta x}{L}, \quad \langle \Delta E \rangle = \langle 2mu(u - v) \rangle = 2mu^2, \quad \langle W \rangle \simeq 2N_s m \frac{\Delta x}{L} u^2 = 10.0 u^2. \quad (7)$$

The scaling is u^2 because Δx is fixed; the u^3 of the plan was the fixed-duration protocol ($\Delta x = u\tau$ grows with u), i.e. Lua–Grosberg’s protocol, not ours. The sign convention is also opposite: their W is done by the gas on the piston (expansion), ours by the piston on the gas.

[INFERENCE] So Au^2 is the natural form at both ends but with different prefactors: $A \approx 25$ in the acoustic regime $u \ll c_s$ (measured up to $u = 0.2$) and $A \rightarrow 2N_s m \Delta x / L = 10$ in the single-hit regime $u \gg v_{\text{th}}$. Between them $A(u)$ must fall, which is the crossover the plan asked for. **[OPEN]** Not yet measured: the fast runs $u = 3, 5, 10$ with travel 1σ (100 seeds, seconds each) test Eq. (7) with no free parameter, and the variance $\text{Var } W \simeq N_{\text{hit}}[4mu^2 k_B T + (2mu^2)^2]$ alongside.

4.3 Jarzynski, kept for later, with its exact requirement

[**SOURCE Jarzynski PRL p. 3**] The ensemble is “microscopic initial conditions for the system and reservoir generated from a thermal equilibrium ensemble at temperature T ”; [**SOURCE pp. 4–5, Eqs. 6–8**] the proof for an isolated system during switching needs only canonical initial conditions and Liouville’s theorem. Consequences for us: (i) canonical seeding is required, drift-first fixed-energy seeding is not it (that is the canonical-seeder flag); (ii) ΔF is the *isothermal* free-energy difference at T_i , $\Delta F = N_s k_B T_i \int_{\eta_i}^{\eta_f} Z/\eta d\eta \approx 6.6$ (bulk KR) — not the adiabatic $W_{\text{qs}} = N_s k_B T_i (e^{\int Z/\eta d\eta} - 1) = 7.07$; the two differ because the adiabat heats the gas; (iii) for compression the rare events that carry $\overline{e^{-\beta W}}$ are the *low-work* trajectories (few hits, hits on receding particles), not Lua–Grosberg’s tail of fast molecules, so their $\exp[mL^2/\tau^2 k_B T]$ estimate does not apply; the cost is set by $\text{Var}(\beta W)$ instead. [**OPEN**] until the canonical seeder exists.

5 The next three moves, in order

1. Ramp against step (box above). Cheap, decides what A is.
2. Paper 1 linewidth against Mansour’s $Q = 1/\mu$ (Section 2.5). Analysis only. If it holds it gives Γ , hence ζ_{hyd} , from data already published in Paper 1.
3. Fast end $u = 3, 5, 10$, travel 1σ , against Eq. (7). Seconds of machine time.

Level 2B (long or absorbing compartment) only after these; it asks whether the recurrence disappears and ζ climbs from ζ_{hyd} toward ζ_0 as the box opens, which is now a prediction, not a fishing trip.

6 The proof ladder, with the source under each arrow

Arrow	Validated by	Status
conservation $\rightarrow$ quasi-static work	first law, $W = \Delta E$; Level 0 ledgers	closed [DATA]
quasi-static work $\rightarrow$ excess energy	W_{qs} from the finite-box adiabat $dE = -PdA$ with measured Z_{wall}	closed at 1.8σ [DATA]
excess energy: its size	$E_{\text{flow}} = N_s m u^2 / 6$ (Mansour Eq. 15/18); acoustic launch $\sim N_s m u^2$	$A = 25.2$ measured; ramp test open
excess energy: no linear term	$\zeta_{\text{hyd}} = L_y \Gamma / X_p$ (Mansour Eq. 17) $\Rightarrow$ slope 0.35	consistent, below resolution
excess energy $\rightarrow$ spring capture	exergy $F_{\text{neq}} - F_{\text{eq}}$ is the capturable part (S–C 150601 p. 1)	Level 3, not started
spring $\rightarrow$ divider transmission	fast pressure channel vs slow heat channel (Mansour p. 1; Crosignani p. 2)	Level 4, not started
transmission $\rightarrow$ multi-wall efficiency	impedance matching $4m_1 m_2 / (m_1 + m_2)^2$ chain	Level 5, not started
(later) Jarzynski	canonical seeding, isolated switching (Jarzynski Eqs. 6–8)	needs the seeder flag